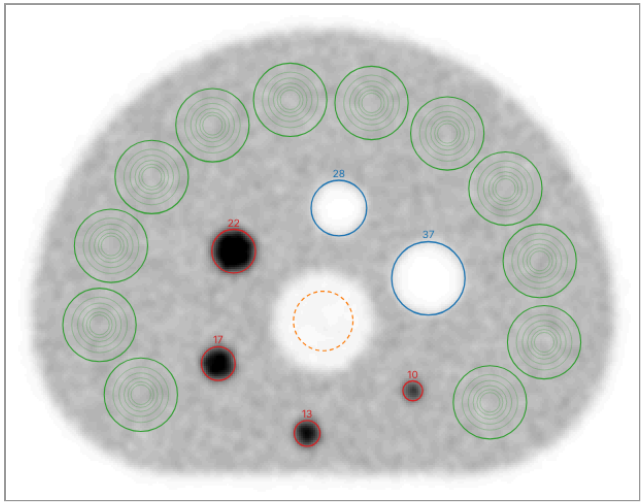
NEMA IQ – Summary (OSEM 3i21s, mean of 3 replicates)

Overall: NO LIMITS

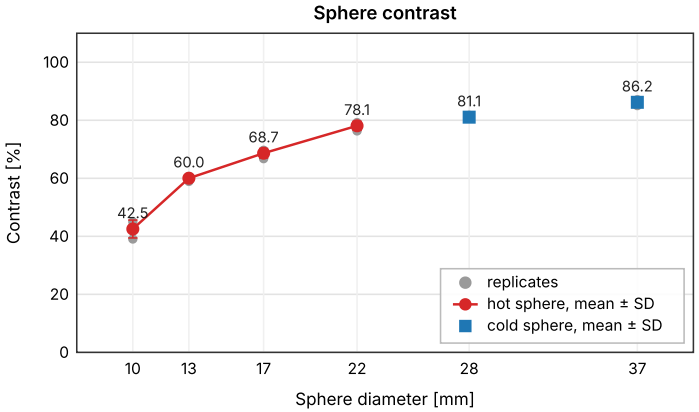
Digital IEC phantom (tests/digital_phantom.py) | Synthetic Digital IEC phantom | study 2026-01-01 | ID DIGITAL PHANTOM
Recon OSEM 3i21s (3 it / 21 ss) | 384x384 | FOV 39.9 cm | pixel 1.040 mm | slice 2.07 mm | dose 72.68 MBq @ 09:00:00 | vol 9358.9 mL | aH/aB = 4.00
Analysis standard: **NEMA NU 2-2012** · limits: No acceptance limits · report generated 2026-10-14 09:57 · NEMA IQ Analyzer v2.0.2

Sphere fill matches NEMA NU 2-2012 (10-22 mm hot, 28/37 mm cold).

Replicate mean, common central slice 48
hot cold background lung

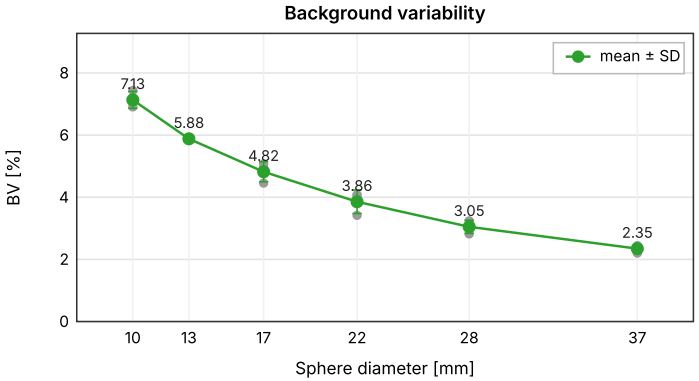


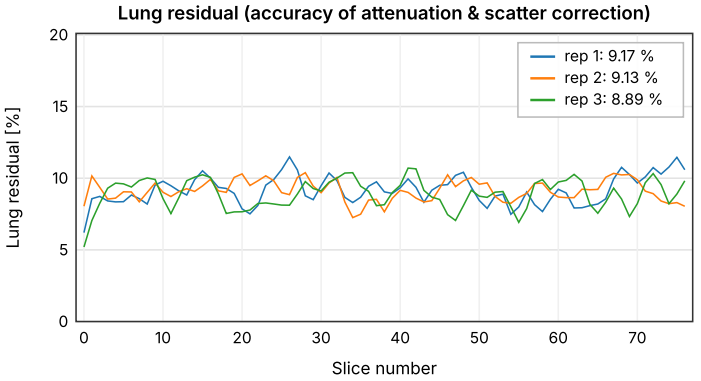
D [mm]	Q rep 1	Q rep 2	Q rep 3	BV rep 1	BV rep 2	BV rep 3
10	43.5	39.1	45.0	7.06	7.44	6.91
13	60.5	60.6	58.9	5.84	5.90	5.90
17	66.8	69.7	69.5	4.93	4.46	5.07
22	76.3	79.2	78.8	4.07	3.42	4.08
28	80.8	81.7	80.7	3.09	2.82	3.24
37	86.4	85.1	87.1	2.20	2.39	2.44
lung	9.17	9.13	8.89			
$C_{B,37}$	5.327	5.067	4.862	kBq/mL		



D [mm]	type	Contrast [%]	limit		BV [%]	limit	
10	hot	42.5 ± 3.1	-	N/A	7.13 ± 0.27	-	N/A
13	hot	60.0 ± 1.0	-	N/A	5.88 ± 0.04	-	N/A
17	hot	68.7 ± 1.6	-	N/A	4.82 ± 0.32	-	N/A
22	hot	78.1 ± 1.5	-	N/A	3.86 ± 0.38	-	N/A
28	cold	81.1 ± 0.6	-	N/A	3.05 ± 0.21	-	N/A
37	cold	86.2 ± 1.0	-	N/A	2.35 ± 0.13	-	N/A
Lung error		9.06 ± 0.15	-	N/A			

Contrast ≥ limit, BV ≤ limit, lung error ≤ limit; limits: No acceptance limits; contrast limits apply to hot spheres. **No acceptance limits entered: no PASS/FAIL.** N/E = limit applies but the sphere was not hot (not evaluable); N/A = no limit.





NEMA IQ – QA details

Overall: NO LIMITS

Digital IEC phantom (tests/digital_phantom.py) | Synthetic Digital IEC phantom | study 2026-01-01 | ID DIGITAL PHANTOM
Recon OSEM 3i21s (3 it / 21 ss) | 384x384 | FOV 39.9 cm | pixel 1.040 mm | slice 2.07 mm | dose 72.68 MBq @ 09:00:00 | vol 9358.9 mL | aH/aB = 4.00
Analysis standard: **NEMA NU 2-2012** · limits: No acceptance limits · report generated 2026-10-14 09:57 · NEMA IQ Analyzer v2.0.2

Protocol item (no published protocol selected)	expected	found	status
Scan time replicate 1 [s]	-	240	INFO
Scan time replicate 2 [s]	-	240	INFO
Scan time replicate 3 [s]	-	240	INFO
Number of replicates	-	3	INFO
Implied bed step (replicate 1), NEMA 100 cm / 30 min [cm]	-	13.3	INFO
Recon method	-	OSEM 3i21s	INFO
Image size	-	384×384	INFO
Pixel size [mm]	-	1.04	INFO
Recon FOV [cm]	-	39.9	INFO
Iterations	-	3	INFO
Subsets	-	21	INFO
Post-filter FWHM [mm]	-	?	INFO
Attenuation corrected	ATTN	yes	OK
Scatter corrected	SCAT	yes	OK
Units	BQML	BQML	OK
Decay correction	START / ADMIN	START	OK

Acquisition

rep 1: IEC digital rep1 — start 10:00:00, 240 s
rep 2: IEC digital rep2 — start 10:06:40, 240 s
rep 3: IEC digital rep3 — start 10:13:20, 240 s

Activity concentration (background, 37 mm ROIs)

rep	$C_{B,37}$ measured	expected	meas/exp
1	5.327	5.300	1.005
2	5.067	5.082	0.997
3	4.862	4.872	0.998

kBq/mL, each replicate decay-corrected to its own start. Expected = net dose 72.68 MBq at 2026-01-01 09:00:00 / volume 9358.9 mL . NEMA NU 2 recommended background at the start of the test: 5.3 kBq/mL (spheres 21.2 kBq/mL at a_H/a_B = 4.00). Replicate consistency after decay correction: CV 0.43 %.

Measured sphere/background ratio

D	type	C_S/C_B rep 1	C_S/C_B rep 2	C_S/C_B rep 3
10	hot	2.30	2.17	2.35
13	hot	2.82	2.82	2.77
17	hot	3.00	3.09	3.09
22	hot	3.29	3.38	3.36
28	cold	0.19	0.18	0.19
37	cold	0.14	0.15	0.13

CTAC check

No matching CT series.

Geometry (automatic localisation)

Common central slice	48 (z = 1.0 mm)
Sphere centres (slice)	10: 47.7, 13: 47.7, 17: 47.6, 22: 47.3, 28: 47.4, 37: 47.3
Sphere ROI slices	10: 48, 13: 48, 17: 48, 22: 48, 28: 48, 37: 48 (common mode)
Sphere ring	radius 57.3 mm (57.2), rms residual 0.12 mm
Tilt / axial spread	lung axis 0.50°, sphere centres 1.0 mm
Background ROIs	margin 15.0 mm (NEMA 15), wall gap 15.0–15.0 mm, sphere gap ≥ 15.1 mm, spacing 40.3 mm
Background slices	38, 43, 48, 53, 58 (-20.7, -10.3, +0.0, +10.3, +20.7 mm)
Lung slices	0–76 (n = 77); insert 0.0–81.5; exclusion 10 mm at the insert end; insert continues beyond the FOV: slices up to the FOV edge used

Standard

	NEMA NU 2-2012	NEMA NU 2-2018
Hot spheres	10, 13, 17, 22 mm	all six (10–37 mm)
Cold spheres	28, 37 mm (Q_C)	none
Lung slices excluded	10 mm from insert ends	30 mm from insert ends
Used here	✔	

Warnings / notes

none

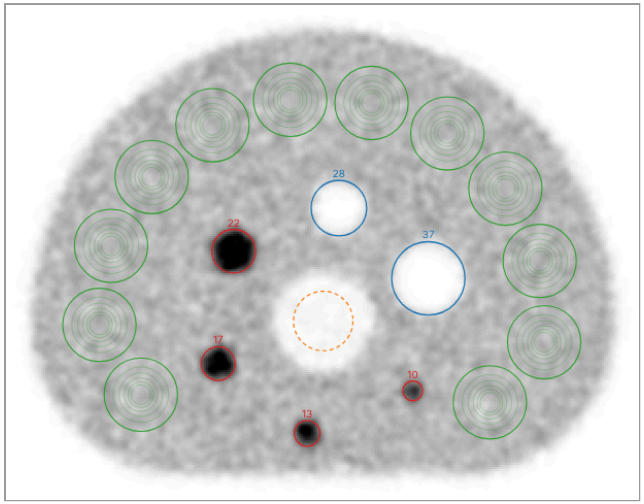
Definitions (NEMA NU 2, Sec. 7)

$Q_H = (C_H/C_B - 1)/(a_H/a_B - 1) \times 100$ %; $Q_C = (1 - C_C/C_B) \times 100$ %; BV = $SD_B/C_B \times 100$ % over 60 background ROIs (12 ROIs × 5 slices, K−1); lung error = $C_{lung,i}/C_{B,37} \times 100$ %, averaged over slices. ROI means use fractional (sub-pixel) weights. a_H/a_B = 4.000.

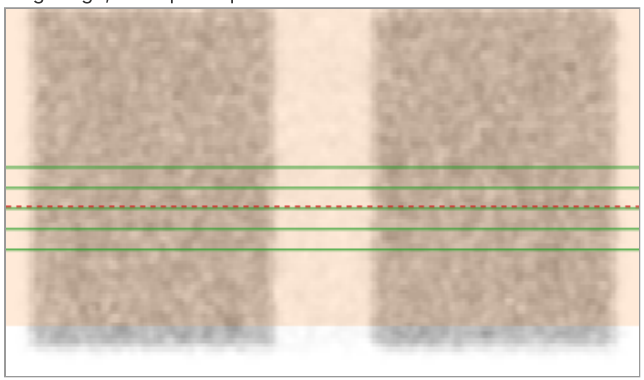
NEMA IQ – Replicate 1/3: IEC digital rep1 (start 10:00:00, 240 s)

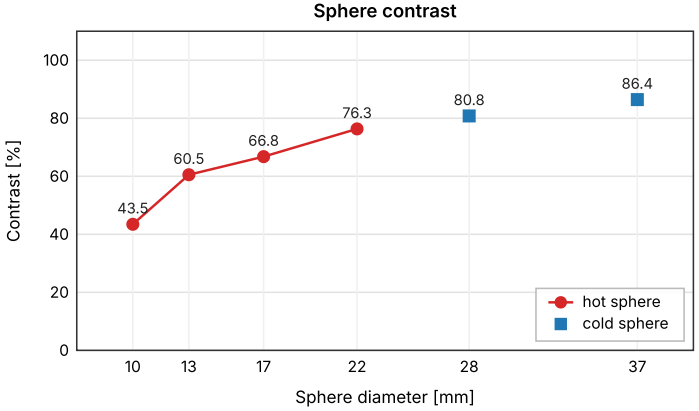
Digital IEC phantom (tests/digital_phantom.py) | Synthetic Digital IEC phantom | study 2026-01-01 | ID DIGITAL PHANTOM
Recon OSEM 3i21s (3 it / 21 ss) | 384x384 | FOV 39.9 cm | pixel 1.040 mm | slice 2.07 mm | dose 72.68 MBq @ 09:00:00 | vol 9358.9 mL | aH/aB = 4.00
Analysis standard: **NEMA NU 2-2012** · limits: No acceptance limits · report generated 2026-10-14 09:57 · NEMA IQ Analyzer v2.0.2

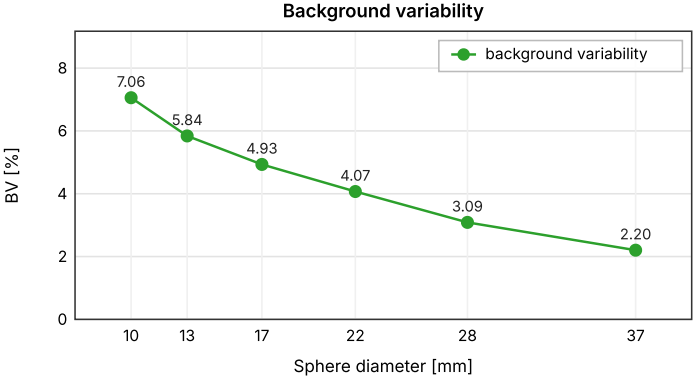
Common central slice 48; sphere ROI slices 10:48, 13:48, 17:48, 22:48, 28:48, 37:48

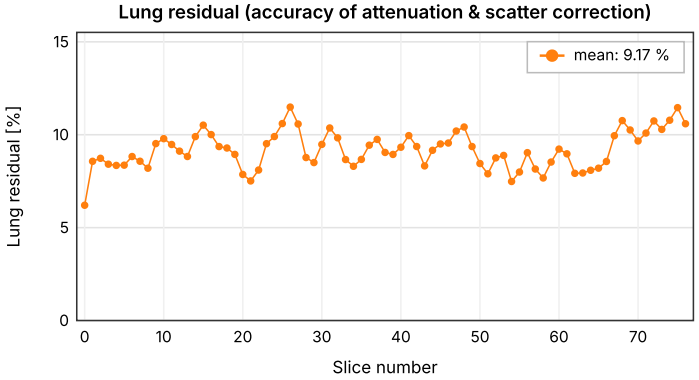


Coronal through lung axis — green: background slices, orange: lung range, red: sphere plane







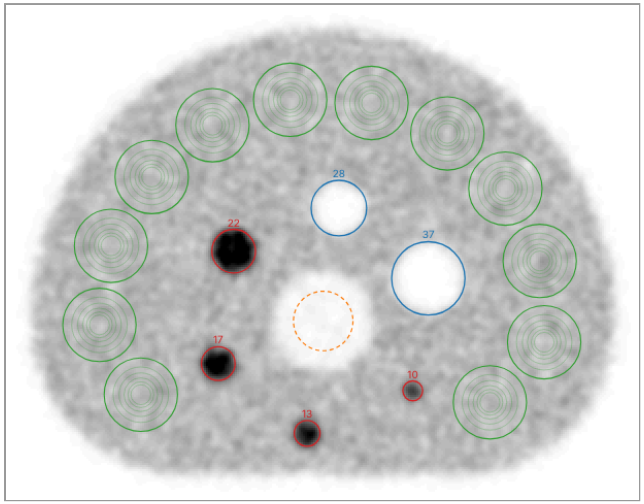


D	type	slice	Q [%]	BV [%]	C_S [kBq/mL]	C_B [kBq/mL]
10	hot	48	43.5	7.06	12.071	5.240
13	hot	48	60.5	5.84	14.797	5.255
17	hot	48	66.8	4.93	15.834	5.272
22	hot	48	76.3	4.07	17.403	5.290
28	cold	48	80.8	3.09	1.020	5.312
37	cold	48	86.4	2.20	0.723	5.327
Lung error			9.17 % (max 11.49 %)			
$C_{B,37}$			5.327 kBq/mL (expected 5.300, ratio 1.005)			

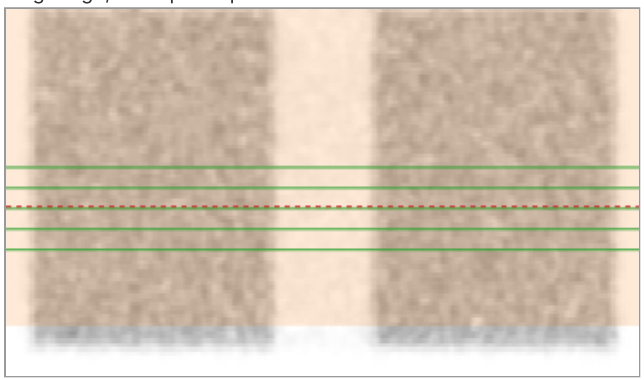
NEMA IQ – Replicate 2/3: IEC digital rep2 (start 10:06:40, 240 s)

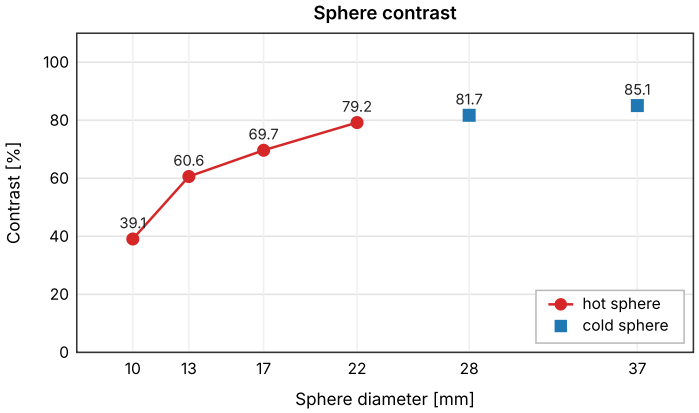
Digital IEC phantom (tests/digital_phantom.py) | Synthetic Digital IEC phantom | study 2026-01-01 | ID DIGITAL PHANTOM
Recon OSEM 3i21s (3 it / 21 ss) | 384x384 | FOV 39.9 cm | pixel 1.040 mm | slice 2.07 mm | dose 72.68 MBq @ 09:00:00 | vol 9358.9 mL | aH/aB = 4.00
Analysis standard: **NEMA NU 2-2012** · limits: No acceptance limits · report generated 2026-10-14 09:57 · NEMA IQ Analyzer v2.0.2

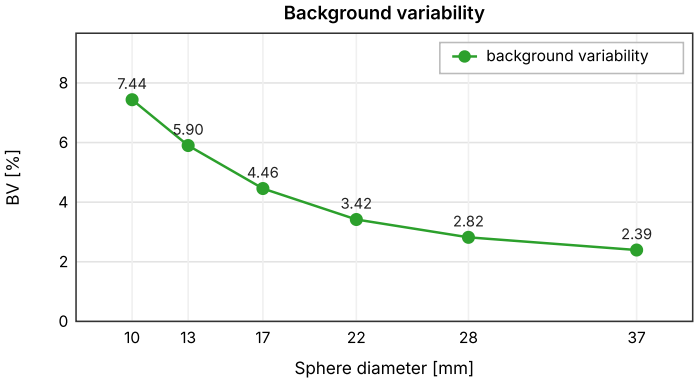
Common central slice 48; sphere ROI slices 10:48, 13:48, 17:48, 22:48, 28:48, 37:48

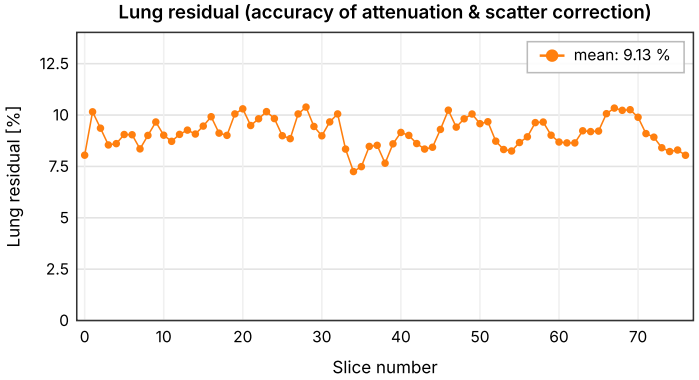


Coronal through lung axis — green: background slices, orange: lung range, red: sphere plane







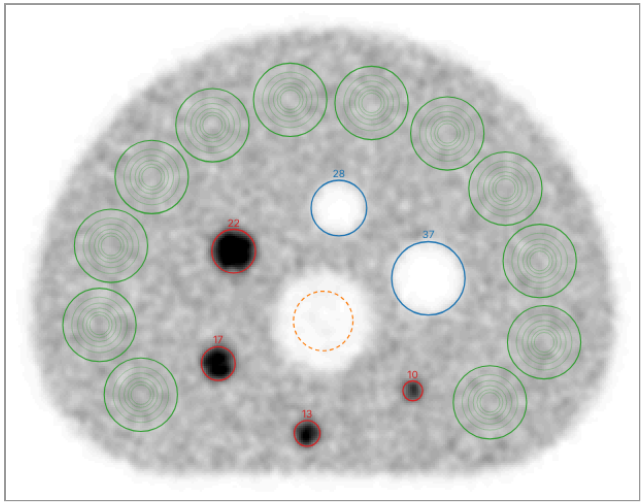


D	type	slice	Q [%]	BV [%]	C_S [kBq/mL]	C_B [kBq/mL]
10	hot	48	39.1	7.44	11.010	5.069
13	hot	48	60.6	5.90	14.297	5.073
17	hot	48	69.7	4.46	15.677	5.074
22	hot	48	79.2	3.42	17.130	5.074
28	cold	48	81.7	2.82	0.927	5.067
37	cold	48	85.1	2.39	0.757	5.067
Lung error			9.13 % (max 10.39 %)			
$C_{B,37}$			5.067 kBq/mL (expected 5.082, ratio 0.997)			

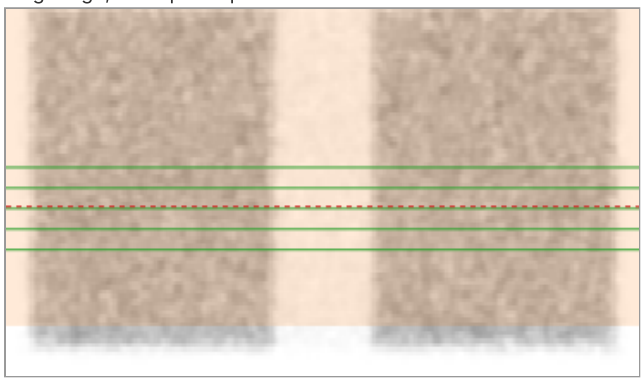
NEMA IQ – Replicate 3/3: IEC digital rep3 (start 10:13:20, 240 s)

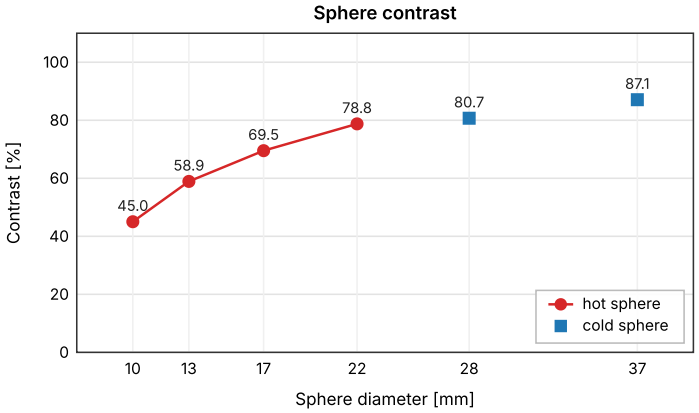
Digital IEC phantom (tests/digital_phantom.py) | Synthetic Digital IEC phantom | study 2026-01-01 | ID DIGITAL PHANTOM
Recon OSEM 3i21s (3 it / 21 ss) | 384x384 | FOV 39.9 cm | pixel 1.040 mm | slice 2.07 mm | dose 72.68 MBq @ 09:00:00 | vol 9358.9 mL | aH/aB = 4.00
Analysis standard: **NEMA NU 2-2012** · limits: No acceptance limits · report generated 2026-10-14 09:57 · NEMA IQ Analyzer v2.0.2

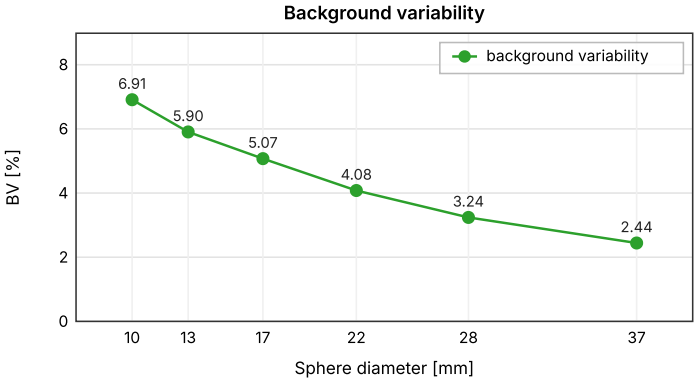
Common central slice 48; sphere ROI slices 10:48, 13:48, 17:48, 22:48, 28:48, 37:48

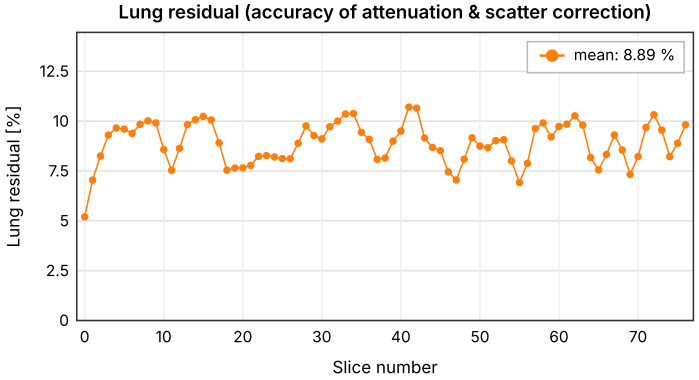


Coronal through lung axis — green: background slices, orange: lung range, red: sphere plane









D	type	slice	Q [%]	BV [%]	C_S [kBq/mL]	C_B [kBq/mL]
10	hot	48	45.0	6.91	11.378	4.840
13	hot	48	58.9	5.90	13.400	4.842
17	hot	48	69.5	5.07	14.938	4.841
22	hot	48	78.8	4.08	16.315	4.852
28	cold	48	80.7	3.24	0.939	4.865
37	cold	48	87.1	2.44	0.628	4.862
Lung error			8.89 % (max 10.71 %)			
$C_{B,37}$			4.862 kBq/mL (expected 4.872, ratio 0.998)			

NEMA IQ – Appendix: ROI data and lung residual per slice

Digital IEC phantom (tests/digital_phantom.py) | Synthetic Digital IEC phantom | study 2026-01-01 | ID DIGITAL PHANTOM
Recon OSEM 3i21s (3 it / 21 ss) | 384x384 | FOV 39.9 cm | pixel 1.040 mm | slice 2.07 mm | dose 72.68 MBq @ 09:00:00 | vol 9358.9 mL | aH/aB = 4.00
Analysis standard: **NEMA NU 2-2012** · limits: No acceptance limits · report generated 2026-10-14 09:57 · NEMA IQ Analyzer v2.0.2

Sphere ROIs (ROI diameter = sphere inner diameter)

D	type	slice	x, y [px]	C_S rep 1	C_S rep 2	C_S rep 3	C_B rep 1	C_B rep 2	C_B rep 3
10	hot	48	238.8, 219.6	12.071	11.010	11.378	5.240	5.069	4.840
13	hot	48	187.6, 240.2	14.797	14.297	13.400	5.255	5.073	4.842
17	hot	48	144.4, 206.4	15.834	15.677	14.938	5.272	5.074	4.841
22	hot	48	151.9, 151.8	17.403	17.130	16.315	5.290	5.074	4.852
28	cold	48	203.0, 131.0	1.020	0.927	0.939	5.312	5.067	4.865
37	cold	48	246.3, 165.0	0.723	0.757	0.628	5.327	5.067	4.862

Concentrations in kBq/mL. C_B = mean of the 60 background ROIs of the sphere's diameter.

Background ROI centres

#	x, y [px]	wall gap [mm]	sphere gap [mm]
1	218.8, 80.0	15.0	23.0
2	255.6, 94.8	15.0	33.9
3	283.7, 121.5	15.0	22.7
4	300.3, 156.6	15.0	19.9
5	302.5, 196.0	15.0	29.7
6	276.2, 225.1	15.0	15.9
7	106.9, 221.4	15.0	15.1
8	86.8, 187.6	15.0	36.0
9	92.4, 149.3	15.0	32.4
10	112.2, 115.9	15.0	26.2
11	141.7, 90.8	15.0	34.9
12	179.3, 78.5	15.0	27.4

Background slices 38, 43, 48, 53, 58.

Lung residual per slice [%] (30 mm ROI / $C_{B,37}$)

slice	z [mm]	rep 1	rep 2	rep 3
0	-98.3	6.20	8.05	5.20
1	-96.3	8.57	10.16	7.04
2	-94.2	8.73	9.36	8.25
3	-92.1	8.41	8.55	9.30
4	-90.0	8.35	8.61	9.66
5	-88.0	8.36	9.05	9.61
6	-85.9	8.83	9.04	9.39
7	-83.8	8.57	8.35	9.84
8	-81.8	8.20	9.01	10.02
9	-79.7	9.52	9.66	9.91
10	-77.6	9.79	9.02	8.58
11	-75.6	9.47	8.72	7.53
12	-73.5	9.12	9.06	8.64
13	-71.4	8.83	9.27	9.83
14	-69.3	9.90	9.08	10.07
15	-67.3	10.52	9.46	10.24
16	-65.2	10.01	9.92	10.06
17	-63.1	9.36	9.12	8.91
18	-61.1	9.29	9.01	7.54
19	-59.0	8.94	10.06	7.65
20	-56.9	7.86	10.30	7.66
21	-54.9	7.51	9.49	7.78
22	-52.8	8.10	9.82	8.24
23	-50.7	9.52	10.17	8.27
24	-48.6	9.90	9.83	8.20
25	-46.6	10.60	8.99	8.12
26	-44.5	11.49	8.85	8.12
27	-42.4	10.58	10.05	8.90
28	-40.4	8.77	10.39	9.77
29	-38.3	8.50	9.44	9.28
30	-36.2	9.48	8.99	9.11
31	-34.2	10.36	9.67	9.72
32	-32.1	9.83	10.05	10.00
33	-30.0	8.66	8.34	10.36
34	-27.9	8.30	7.25	10.39
35	-25.9	8.67	7.49	9.44
36	-23.8	9.44	8.47	9.09
37	-21.7	9.75	8.53	8.08
38	-19.7	9.05	7.66	8.15

slice	z [mm]	rep 1	rep 2	rep 3
39	-17.6	8.94	8.60	9.00
40	-15.5	9.33	9.15	9.50
41	-13.5	9.95	9.01	10.71
42	-11.4	9.37	8.61	10.65
43	-9.3	8.32	8.34	9.15
44	-7.2	9.16	8.43	8.68
45	-5.2	9.50	9.30	8.52
46	-3.1	9.55	10.24	7.46
47	-1.0	10.20	9.41	7.05
48	1.0	10.42	9.82	8.09
49	3.1	9.36	10.05	9.17
50	5.2	8.45	9.58	8.75
51	7.2	7.90	9.67	8.66
52	9.3	8.75	8.73	9.02
53	11.4	8.89	8.32	9.07
54	13.5	7.48	8.25	8.01
55	15.5	7.99	8.66	6.92
56	17.6	9.04	8.94	7.88
57	19.7	8.15	9.63	9.63
58	21.7	7.67	9.66	9.91
59	23.8	8.53	9.02	9.21
60	25.9	9.23	8.69	9.73
61	27.9	8.97	8.65	9.85
62	30.0	7.92	8.64	10.27
63	32.1	7.94	9.23	9.80
64	34.2	8.09	9.20	8.17
65	36.2	8.20	9.22	7.56
66	38.3	8.56	10.07	8.33
67	40.4	9.95	10.34	9.31
68	42.4	10.76	10.23	8.55
69	44.5	10.26	10.26	7.33
70	46.6	9.66	9.89	8.23
71	48.6	10.09	9.10	9.68
72	50.7	10.74	8.92	10.32
73	52.8	10.29	8.41	9.55
74	54.9	10.78	8.22	8.22
75	56.9	11.46	8.30	8.89
76	59.0	10.59	8.04	9.82